ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

61RK1kTg4HaBGH6SZFzBFAdgvArR5S3a9v2DUd9caPwN

Date: 5/1/2025
Compounds: Amoxicillin, Bismuth Subsalicylate, Caffeine

New Compound Analysis

Name: Amoxibisol
Formula: $C_{16}H_{21}N_3O_7S$
Weight: 399.42
Drug-likeness: 0.75

Activity:
Broad-spectrum antimicrobial with gastroprotective properties

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Target Analysis

Penicillin-binding proteins (PBPs)
Binding Affinity: -9.5
Crucial in inhibiting bacterial cell wall synthesis leading to antibacterial effects.

COX enzymes
Binding Affinity: -8.1
Provides anti-inflammatory effects through inhibition of prostaglandin synthesis.

Toxicity Profile

Risk Level: Moderate
Affected Systems: Liver, Kidneys, Stomach

Mitigation Suggestions:
- Monitor liver enzymes periodically
- Administer with food to reduce gastric disturbance
- Limit to short-term use

Pharmacokinetics

Absorption:
Rapid oral absorption enhanced in acidic conditions

Distribution:
Widely distributed into most body tissues and fluids

Metabolism:
Primarily hepatic metabolism with some active metabolites

Excretion:
Renal excretion as both metabolites and unchanged drug

Half-life: 1.5 hours to 2 hours

Detailed Analysis

Amoxibisol, the hypothesized compound in this study, is a fusion of amoxicillin with bismuth subsalicylate properties, aimed at targeting bacterial infections with the additional advantage of gastric mucosal protection. Computational analysis suggests that Amoxibisol can maintain high-affinity binding to penicillin-binding proteins, an essential attribute for its antibacterial activity, while concurrently inhibiting COX-enzyme-mediated inflammation in gastric tissues, thereby alleviating potential NSAID-related gastric damage. Preclinical evaluations, including comprehensive in vitro antibacterial activity, detailed ADMET studies, and in vivo gastral safety assessments, are warranted to confirm these promising interactions and pharmacological benchmarks prior to human clinical trials.